

5.5.2 Substitution w/ Definite Integrals Worksheet

Evaluate each of the following integrals.

1. $\int_0^1 \sqrt[3]{1+7x} dx$

Let $u = 1 + 7x$ and $du = 7dx \rightarrow \frac{1}{7} du = dx$.

The bounds become $u(0) = 1 + 7(0) = 1$ & $u(1) = 1 + 7(1) = 8$.

Therefore,

$$\int_0^1 \sqrt[3]{1+7x} dx = \int_1^8 \frac{1}{7} \sqrt[3]{u} du = \int_1^8 \frac{1}{7} u^{1/3} du = \frac{3}{28} u^{4/3} \Big|_1^8 = \frac{3}{28} (8)^{4/3} - \frac{3}{28} (1)^{4/3} = \frac{3}{28} (16) - \frac{3}{28} = \frac{45}{28}$$

2. $\int_0^{\pi} \sec^2\left(\frac{t}{4}\right) dt$

Let $u = \frac{t}{4}$ and $du = \frac{1}{4} dt \rightarrow 4du = dt$.

The bounds become $u(0) = 0$ & $u(\pi) = \frac{\pi}{4}$.

Therefore,

$$\int_0^{\pi} \sec^2\left(\frac{t}{4}\right) dt = \int_0^{\pi/4} 4 \sec^2(u) du = 4 \tan(u) \Big|_0^{\pi/4} = 4 \tan\left(\frac{\pi}{4}\right) - 4 \tan(0) = 4(1) - 4(0) = 4$$

3. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$

Let $u = 1 + 2x$ and $du = 2dx \rightarrow \frac{1}{2} du = dx$.

The bounds become $u(0) = 1 + 2(0) = 1$ & $u(13) = 1 + 2(13) = 27$.

Therefore,

$$\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}} = \int_1^{27} \frac{1}{7} u^{-2/3} du = \frac{1}{21} u^{1/3} \Big|_1^{27} = \frac{1}{21} (3)^{1/3} - \frac{1}{21} (1)^{1/3} = \frac{3}{21} - \frac{1}{21} = \frac{2}{21}$$

$$4. \int_1^2 x\sqrt{x-1}dx$$

Let $u = x - 1$ and $du = dx$. We also know that $x = u + 1$.

The bounds become $u(1) = (1) - 1 = 0$ & $u(2) = (2) - 1 = 1$.

Therefore,

$$\int_1^2 x\sqrt{x-1}dx = \int_0^1 (u+1)u^{1/2}du = \int_0^1 (u^{3/2} + u^{1/2})du = \left(\frac{2}{5}u^{5/2} + \frac{2}{3}u^{3/2} \right) \Big|_0^1 = \left(\frac{2}{5} + \frac{2}{3} \right) - (0+0) = \left(\frac{6}{15} + \frac{10}{15} \right) = \frac{16}{15}$$

$$5. \int_{1/2}^1 \frac{\cos(x^{-2})}{x^3} dx$$

Let $u = x^{-2}$ and $du = -2x^{-3}dx \rightarrow -\frac{1}{2}du = \frac{1}{x^3}dx$.

The bounds become $u\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{-2} = 4$ & $u(1) = (1)^{-2} = 1$.

Therefore,

$$\int_{1/2}^1 \frac{\cos(x^{-2})}{x^3} dx = \int_4^1 -\frac{\cos(u)}{2} du = \left(-\frac{1}{2}\sin(u) \right) \Big|_4^1 = -\frac{1}{2}\sin(1) + \frac{1}{2}\sin(4)$$

$$6. \int_0^{\sqrt{\pi}} x \cos(x^2) dx$$

Let $u = x^2$ and $du = 2xdx \rightarrow \frac{1}{2}du = xdx$.

The bounds become $u(0) = 0$ & $u(\sqrt{\pi}) = \pi$.

Therefore,

$$\int_0^{\sqrt{\pi}} x \cos(x^2) dx = \int_0^{\pi} \frac{1}{2} \cos(u) du = \left(\frac{1}{2} \sin(u) \right) \Big|_0^{\pi} = \frac{1}{2} \sin(\pi) - \frac{1}{2} \sin(0) = 0 - 0 = 0.$$

$$7. \int_{1/6}^{1/2} \csc(\pi t) \cot(\pi t) dt$$

Let $u = \pi t$ and $du = \pi dt \rightarrow \frac{1}{\pi} du = dt$.

The bounds become $u\left(\frac{1}{6}\right) = \frac{\pi}{6}$ & $u\left(\frac{1}{2}\right) = \frac{\pi}{2}$.

Therefore,

$$\begin{aligned} \int_{1/6}^{1/2} \csc(\pi t) \cot(\pi t) dt &= \int_{\pi/6}^{\pi/2} \frac{1}{\pi} \csc(u) \cot(u) du = \left(-\frac{1}{\pi} \csc(u) \right) \Big|_{\pi/6}^{\pi/2} = -\frac{1}{\pi} \csc\left(\frac{\pi}{2}\right) + \frac{1}{\pi} \csc\left(\frac{\pi}{6}\right) \\ &= -\frac{1}{\pi}(1) + \frac{1}{\pi}(2) = \frac{1}{\pi} \end{aligned}$$

$$8. \int_0^{\pi/2} \cos(x) \sin(\sin(x)) dx$$

Let $u = \sin(x)$ and $du = \cos(x) dx$.

The bounds become $u(0) = \sin(0) = 0$ & $u\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$.

Therefore,

$$\int_0^{\pi/2} \cos(x) \sin(\sin(x)) dx = \int_0^1 \sin(u) du = -\cos(u) \Big|_0^1 = -\cos(1) + \cos(0) = 1 - \cos(1).$$

$$9. \int_0^4 \frac{x}{\sqrt{1+2x}} dx$$

Let $u = 1 + 2x$ and $du = 2dx \rightarrow \frac{1}{2} du = dx$. Also $x = \frac{1}{2}(u - 1)$.

The bounds become $u(0) = 1 + 2(0) = 1$ & $u(4) = 1 + 2(4) = 9$.

Therefore,

$$\int_0^4 \frac{x}{\sqrt{1+2x}} dx = \int_1^9 \frac{1}{4} \frac{(u-1)}{\sqrt{u}} du = \int_1^9 \frac{1}{4} (u^{1/2} - u^{-1/2}) du = \frac{1}{4} \left(\frac{2}{3} u^{3/2} - 2u^{1/2} \right) \Big|_1^9 = \frac{1}{6} (9)^{3/2} - \frac{1}{2} (1)^{3/2} = \frac{27}{6} - \frac{1}{2} = \frac{15}{6} = \frac{5}{2}$$

$$10. \int_e^{e^4} \frac{dx}{x\sqrt{\ln(x)}}$$

Let $u = \ln(x)$ and $du = \frac{1}{x} dx$.

The bounds become $u(e) = \ln(e) = 1$ & $u(e^4) = \ln(e^4) = 4$.

Therefore,

$$\int_e^{e^4} \frac{dx}{x\sqrt{\ln(x)}} = \int_1^4 \frac{1}{\sqrt{u}} du = \int_1^4 u^{-1/2} du = 2u^{1/2} \Big|_1^4 = 2(4)^{1/2} - 2(1)^{1/2} = 2(2) - 2(1) = 3.$$

$$11. \int_0^1 \frac{e^z + 1}{e^z + z} dz$$

Let $u = e^z + z$ and $du = (e^z + 1) dz$.

The bounds become $u(0) = e^0 + 0 = 1$ & $u(1) = e^1 + 1 = e + 1$.

Therefore,

$$\int_0^1 \frac{e^z + 1}{e^z + z} dz = \int_1^{e+1} \frac{1}{u} du = \ln|u| \Big|_1^{e+1} = \ln|e+1| - \ln|1| = \ln|e+1|.$$

$$12. \int_0^1 x e^{-x^2} dx$$

Let $u = -x^2$ and $du = -2x dx \rightarrow -\frac{1}{2} du = x dx$.

The bounds become $u(0) = 0$ & $u(1) = -1$.

Therefore,

$$\int_0^1 x e^{-x^2} dx = \int_0^{-1} -\frac{1}{2} e^u du = -\frac{1}{2} e^u \Big|_0^{-1} = -\frac{1}{2} e^{-1} + \frac{1}{2} e^0 = -\frac{1}{2e} + \frac{1}{2}$$

$$13. \int_0^{1/2} \frac{\sin^{-1}(x)}{\sqrt{1-x^2}} dx$$

Let $u = \sin^{-1}(x)$ and $du = \frac{1}{\sqrt{1-x^2}} dx$.

The bounds become $u(0) = \sin^{-1}(0) = 0$ & $u\left(\frac{1}{2}\right) = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$.

Therefore,

$$\int_0^{1/2} \frac{\sin^{-1}(x)}{\sqrt{1-x^2}} dx = \int_0^{\pi/6} u du = \frac{1}{2} u^2 \Big|_0^{\pi/6} = \frac{1}{2} \left(\frac{\pi}{6}\right)^2 - \frac{1}{2} (0)^2 = \frac{\pi^2}{72}$$

$$14. \int_0^1 \frac{dx}{(1+\sqrt{x})^4}. \text{ Hint: Try to rewrite your } du \text{ to be fully in terms of } u \text{ instead of } x.$$

Let $u = 1 + \sqrt{x}$ and thus $\sqrt{x} = u - 1$ and $du = \frac{1}{2\sqrt{x}} dx \rightarrow 2\sqrt{x} du = dx \rightarrow 2(u-1) du = dx$.

The bounds become $u(0) = 1$ & $u(1) = 2$.

Therefore,

$$\begin{aligned} \int_0^1 \frac{dx}{(1+\sqrt{x})^4} &= \int_1^2 \frac{2(u-1)}{u^4} du = \int_1^2 (2u^{-3} - 2u^{-4}) du = \left(-u^{-2} + \frac{2}{3} u^{-3} \right) \Big|_1^2 \\ &= \left[-(2)^{-2} + \frac{2}{3} (2)^{-3} \right] - \left[-(1)^{-2} + \frac{2}{3} (1)^{-3} \right] \\ &= \left[-\frac{1}{4} + \frac{1}{12} \right] - \left[-1 + \frac{2}{3} \right] = \left[-\frac{2}{12} \right] - \left[-\frac{1}{3} \right] = \frac{2}{12} = \frac{1}{6} \end{aligned}$$

#15 – 18 Use properties of even and odd functions to help answer the following questions.

$$15. \text{ Evaluate } \int_{-\pi/3}^{\pi/3} x^4 \sin(x) dx.$$

We note that $f(x) = x^4 \sin(x)$.

Then we see, $f(-x) = (-x)^4 \sin(-x) = -x^4 \sin(x) = -f(x)$. This implies that f is an odd function.

Therefore, $\int_{-\pi/3}^{\pi/3} x^4 \sin(x) dx = 0$.

16. Evaluate $\int_{-1}^1 \left(x^5 - 6x^9 + \frac{\sin(x)}{(1+x^4)^2} \right) dx.$

We note that $f(x) = x^5 - 6x^9 + \frac{\sin(x)}{(1+x^4)^2}.$

Then we see, $f(-x) = (-x)^5 - 6(-x)^9 + \frac{\sin(-x)}{(1+(-x)^4)^2} = -x^5 + 6x^9 - \frac{\sin(x)}{(1+x^4)^2} = -f(x).$ This implies that

f is an odd function.

Therefore, $\int_{-1}^1 \left(x^5 - 6x^9 + \frac{\sin(x)}{(1+x^4)^2} \right) dx = 0.$

17. Evaluate $\int_{-1}^1 \frac{\tan(x)}{1+x^2} dx.$

We note that $f(x) = \frac{\tan(x)}{1+x^2}.$

Then we see, $f(-x) = \frac{\tan(-x)}{1+(-x)^2} = -\frac{\tan(x)}{1+x^2} = -f(x).$ This implies that f is an odd function.

Therefore, $\int_{-1}^1 \frac{\tan(x)}{1+x^2} dx = 0.$

18. Evaluate $\int_{-\pi/4}^{\pi/4} \frac{t^4 \sin(t)}{2+\cos(t)} dt.$

We note that $f(t) = \frac{t^4 \sin(t)}{2+\cos(t)}.$

Then we see, $f(-t) = \frac{(-t)^4 \sin(-t)}{2+\cos(-t)} = -\frac{t^4 \sin(t)}{2+\cos(t)} = -f(t).$ This implies that f is an odd function.

Therefore, $\int_{-\pi/4}^{\pi/4} \frac{t^4 \sin(t)}{2+\cos(t)} dt = 0.$

19. Suppose f is an odd function and g is an even function, each integrable on $[-2, 2]$. Give that

$$\int_0^2 f(x) dx = 3, \quad \int_2^6 f(x) dx = 13 \text{ and } \int_0^2 g(x) dx = 6, \text{ evaluate the following.}$$

a) $\int_{-2}^2 [4f(x) - 5g(x)] dx$

We note that

$$\begin{aligned} \int_{-2}^2 [4f(x) - 5g(x)] dx &= \int_{-2}^2 4f(x) dx - \int_{-2}^2 5g(x) dx = 4 \int_{-2}^2 f(x) dx - 5 \int_{-2}^2 g(x) dx \\ &= 4(0) - 10 \int_0^2 g(x) dx = 0 - 10(6) = -60 \end{aligned}$$

b) $\int_{-2}^2 [3g(x) + x^3 - 7] dx$

We note that $\int_{-2}^2 [3g(x) + x^3 - 7] dx = \int_{-2}^2 3g(x) dx + \int_{-2}^2 x^3 dx - \int_{-2}^2 7 dx$.

Now, since $y = x^3$ is an odd function and $y = 7$ is a constant we get that

$$\int_{-2}^2 3g(x) dx + \int_{-2}^2 x^3 dx - \int_{-2}^2 7 dx = 3 \int_{-2}^2 g(x) dx + 0 - 7(2 - (-2)) = 6 \int_0^2 g(x) dx + 0 - 7(4) = 6(6) + 0 - 28 = 8$$

c) $\int_{-2}^2 [f(x)g(x)] dx$

We note that $h(x) = f(x)g(x)$.

Then we see, $h(-x) = f(-x)g(-x) = -f(x)g(x) = -h(x)$. This implies that h is an odd function.

Therefore, $\int_{-2}^2 [f(x)g(x)] dx = 0$.

d) $\int_{-2}^0 f(x) dx$

Since $f(x)$ is an odd function we know that $\int_{-2}^2 f(x) dx = 0$. We also know that

$$\int_{-2}^2 f(x) dx = \int_{-2}^0 f(x) dx + \int_0^2 f(x) dx.$$

Then we get $0 = \int_{-2}^0 f(x) dx + 3 \rightarrow \int_{-2}^0 f(x) dx = -3$.

$$\text{e) } \int_{-2}^2 \left[(f(x))^3 + 4 \right] dx$$

We note that $h(x) = (f(x))^3$.

Then we see, $h(-x) = (f(-x))^3 = (-f(x))^3 = -(f(x))^3 = -h(x)$. This implies that h is an odd function. Therefore, $\int_{-2}^2 (f(x))^3 dx = 0$.

$$\text{So, we get } \int_{-2}^2 \left[(f(x))^3 + 4 \right] dx = \int_{-2}^2 (f(x))^3 dx + \int_{-2}^2 4 dx = 0 + 4(2 - (-2)) = 16$$

$$\text{f) } \int_{-6}^2 [f(x) + 2] dx$$

Note that

$$\begin{aligned} \int_{-6}^2 [f(x) + 2] dx &= \int_{-6}^2 f(x) dx + \int_{-6}^2 2 dx = \left[\int_{-6}^6 f(x) dx - \int_2^6 f(x) dx \right] + 2(2 - (-6)) \\ &= [0 - 13] + 2(8) = -13 + 16 = 3 \end{aligned}$$

20. If $f'(x)$ is continuous on $[a, b]$, show that $2 \int_a^b f(x) f'(x) dx = [f(b)]^2 - [f(a)]^2$.

We start with $2 \int_a^b f(x) f'(x) dx$.

Let $u = f(x)$ and $du = f'(x) dx$.

The bounds become $u(a) = f(a)$ & $u(b) = f(b)$.

$$2 \int_a^b f(x) f'(x) dx = 2 \int_{f(a)}^{f(b)} u du = u^2 \Big|_{f(a)}^{f(b)} = f(b)^2 - f(a)^2$$

21. If $f(x)$ is continuous on $[0, 1]$, prove that $\int_0^1 f(x) dx = \int_0^1 f(1-x) dx$.

We start with $\int_0^1 f(1-x) dx$.

Let $u = 1-x$ and $du = -1dx \rightarrow -du = dx$.

The bounds become $u(0) = 1$ & $u(1) = 0$.

$$\int_0^1 f(1-x) dx = \int_1^0 -f(u) du = \int_0^1 f(u) du = \int_0^1 f(x) dx$$

22. If $f(x)$ is continuous and $\int_0^4 f(x) dx = 10$, find $\int_0^2 f(2x) dx$.

We start with $\int_0^2 f(2x) dx$.

Let $u = 2x$ and $du = 2dx \rightarrow \frac{1}{2} du = dx$.

The bounds become $u(0) = 0$ & $u(2) = 4$.

$$\int_0^2 f(2x) dx = \int_0^4 \frac{1}{2} f(u) du = \frac{1}{2} \int_0^4 f(u) du = \frac{1}{2} \int_0^4 f(x) dx = \frac{1}{2}(10) = 5$$

23. If $f(x)$ is continuous and $\int_0^9 f(x) dx = 4$, find $\int_0^3 xf(x^2) dx$.

We start with $\int_0^3 xf(x^2) dx$.

Let $u = x^2$ and $du = 2xdx \rightarrow \frac{1}{2} du = xdx$.

The bounds become $u(0) = 0$ & $u(3) = 9$.

$$\int_0^3 xf(x^2) dx = \int_0^9 \frac{1}{2} f(u) du = \frac{1}{2} \int_0^9 f(u) du = \frac{1}{2} \int_0^9 f(x) dx = \frac{1}{2}(4) = 2$$